NILANJAN MANDAL

Backend Engineer | Python, FastAPI, PostgreSQL & AWS

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PROFESSIONAL SUMMARY

Backend-focused full-stack engineer with 2+ years of full-time experience and 2.5 years of internship experience. Builds Python/FastAPI and Node.js services with PostgreSQL, Redis and AWS. Delivered rewards and wallet APIs for four branded apps, improved dashboard performance, and implemented reliable payment, background-job and real-time workflows.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Go
Backend: FastAPI, Node.js, Express, REST APIs, Pydantic, SQLAlchemy, JWT, OAuth 2.0, RBAC
Data & async: PostgreSQL, Redis, Celery, BullMQ, WebSockets, python-socketio
Cloud & delivery: AWS (S3, Lambda, EC2), Docker, GitHub Actions, Git
Testing: pytest, Jest, Supertest, Playwright
Frontend & mobile: React, Next.js, React Native, Expo, Tailwind CSS

PROFESSIONAL EXPERIENCE

Member of Technical Staff | (23 Ventures) Tevis Software Private Limited

Sep 2024 - Oct 2026 | Remote, India

- Designed FastAPI REST services for venues, campaigns, prizes and wallets, consumed by four React Native apps and a React admin console.
- Modelled PostgreSQL schemas with async SQLAlchemy; added indexes and tuned queries to reduce p95 dashboard load time from approximately 800 ms to under 150 ms.
- Implemented JWT access and refresh-token rotation, role-based access control, Redis rate limiting and Pydantic validation across write endpoints.
- Added idempotency keys and transaction-safe wallet ledger entries, eliminating duplicate payouts observed during load testing.
- Moved settlement, notifications and bulk redemption processing to Celery/Redis workers with retries, backoff and dead-letter handling for thousands of jobs per day.
- Scaled python-socketio event delivery across Uvicorn workers using Redis, supporting mission progress, rankings and notifications for thousands of concurrent players.
- Built resumable S3 multipart uploads with boto3, concurrent transfers and presigned-URL renewal; containerised services and added GitHub Actions checks with pytest integration tests against ephemeral PostgreSQL.

Software Engineering Intern | StepOut

Nov 2021 - May 2024 | Remote, India

- Rebuilt Node.js video ingestion with parallel, resumable S3 multipart uploads, increasing reliable upload size from approximately 500 MB to multi-gigabyte files.
- Integrated Razorpay order APIs, payment capture, webhook signature verification and idempotent reconciliation to keep payment records consistent with the gateway.
- Extended PostgreSQL models and migrations for video metadata and multilingual content; built AWS Lambda/Playwright workflows for PDF generation and document merging.

SELECTED PROJECTS

HushMark | PII detection and redaction | hush-mark.nilanjan.xyz

- Built a React/TypeScript app with a FastAPI backend for text, document and audio redaction. Combined OpenAI entity detection and Whisper transcription with human review for low-confidence entities; moved long-running work to async workers and packaged the stack with Docker.

Flixr | YouTube publishing workflow automation

- Built a Next.js and Node.js/Express publishing workflow with PostgreSQL approval states, JWT roles and a Redis/BullMQ upload queue. Integrated YouTube Data API v3 with per-channel OAuth 2.0, encrypted refresh tokens, quota-aware retries and persistent queue state.

EDUCATION

B.Tech in Information Technology | RCC Institute of Information Technology, Kolkata
2024 | CGPA: 8.37